

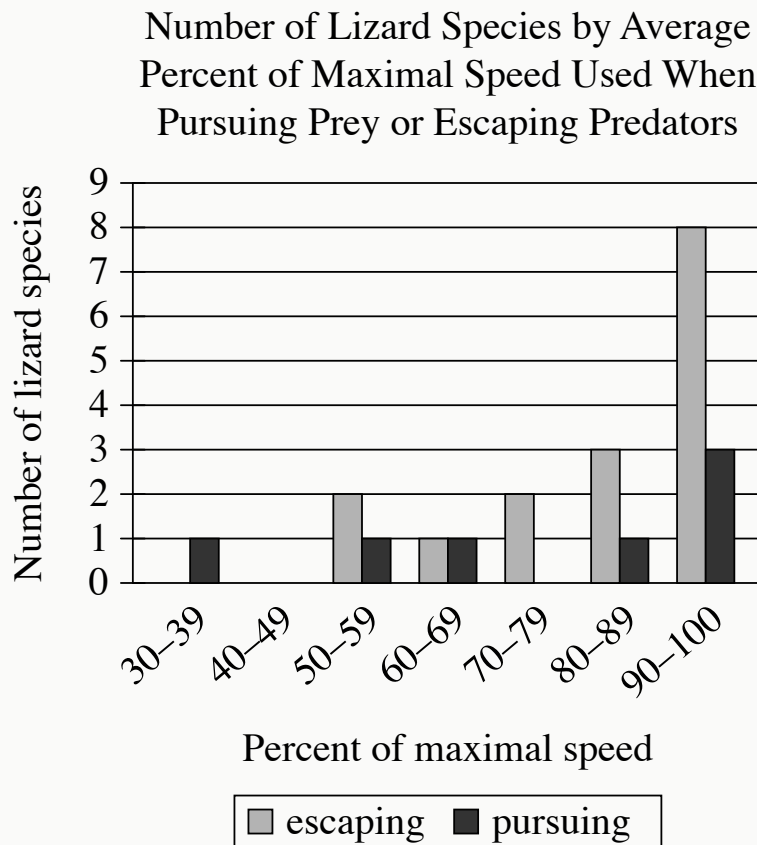
● HARD

● INFORMATION AND IDEAS

Information and Ideas

10

10 questions · ~15 min



- For each data category, the following bars are shown:
 - escaping
 - pursuing
- The data for the 7 categories are as follows:
 - 30–39 percent of maximal speed:
 - escaping: 0 species
 - pursuing: 1 species
 - 40–49 percent of maximal speed:
 - escaping: 0 species
 - pursuing: 0 species
 - 50–59 percent of maximal speed:
 - escaping: 2 species
 - pursuing: 1 species

- 60–69 percent of maximal speed:
 - escaping: 1 species
 - pursuing: 1 species
- 70–79 percent of maximal speed:
 - escaping: 2 species
 - pursuing: 0 species
- 80–89 percent of maximal speed:
 - escaping: 3 species
 - pursuing: 1 species
- 90–100 percent of maximal speed:
 - escaping: 8 species
 - pursuing: 3 species

It may seem that the optimal strategy for an animal pursuing prey or escaping predators is to move at maximal speed, but the energy expense of exploiting full speed capacity can disfavor such a strategy even in escape contexts, as evidenced by the fact that _____ blank

Which choice most effectively uses data from the graph to complete the text?

- A. most lizard species use about the same percentage of their maximal speed when escaping predation as they do when pursuing prey.
- B. multiple lizard species move at an average of less than 90% of their maximal speed while escaping predation.
- C. more lizard species use, on average, 90%–100% of their maximal speed while escaping predation than use any other percentage of their maximal speed.
- D. at least 4 lizard species use, on average, less than 100% of their maximal speed while pursuing prey.

During the Bourbon Restoration in France (1814–1830), the right to vote required in part that a person paid at least 300 francs in direct taxes to the government. The four most common taxes (the *quatre vieilles*) were levied on real estate (both land and buildings); the doors and windows in taxpayer homes; the rental values of homes; and the businesses of artisans and merchants. (Foreign investments were either exempt from taxation or taxed lightly.) Although relatively few people paid the tax on real estate, it was the main means of voter qualification and accounted for over two-thirds of government receipts during this period, suggesting that during the Bourbon Restoration _____ blank

Which choice most logically completes the text?

- A. those people who had the right to vote most likely had substantial holdings of French real estate.
- B. the voting habits of French artisans and merchants were effective in reducing tax burdens on businesses.
- C. the number of doors and windows in French residences was kept to a minimum but increased after 1830.
- D. French people with significant foreign investments were unlikely to have the right to vote.

The fynbos shrubland is a diverse habitat found only in South Africa. It is adjacent to the Afro-temperate forest, with almost no transition space between the two distinct habitats. Plants in the fynbos have uniquely thin and long root systems that spread out over large distances to absorb nutrients from the soil. Ecologists transplanted tree seedlings from the forest into plots in the fynbos. Seedlings in plots isolated from the roots of fynbos plants exhibited a growth rate five times greater than that of the seedlings in plots in close proximity to the roots of fynbos plants.

Based on the text, what role do fynbos roots most likely have in maintaining the border between the fynbos shrubland and the Afro-temperate forest habitats?

- A. Fynbos roots damage the root systems of forest plants, leaving those plants unable to acquire sufficient nutrients.
- B. Fynbos roots extend close enough to the forest plants' roots that they constitute a physical barrier that forest plants' roots cannot pass.
- C. The root systems of fynbos plants allow the plants to take in so many soil nutrients that forest plants are prevented from flourishing in the fynbos.
- D. The root systems of fynbos plants enhance the soil immediately surrounding the plants, allowing them to thrive in an otherwise harsh habitat.

Initially observed in 2017, the interstellar object ‘Oumuamua is the first object of its kind to be seen in our solar system. Researchers have been puzzled because its acceleration cannot be entirely explained by the gravitational pull of nearby bodies: there must be a nongravitational influence on its velocity and trajectory. Some previously suggested explanations for this nongravitational acceleration involve mechanisms that are unlikely or unrealistic, such as geometric effects from ‘Oumuamua being potentially composed of several spatially separated bodies. Now, Jennifer Bergner and colleagues propose that the nongravitational acceleration is due to the gaseous expulsion of entrapped hydrogen from ‘Oumuamua’s water-rich icy body.

Which statement, if true, would most strongly support the claim made by Bergner and colleagues about the cause of ‘Oumuamua’s acceleration?

- A. Existing proposed models of outgassing from ‘Oumuamua include the direct conversion of nitrogen or carbon monoxide from a solid to a gaseous state without becoming liquid, but these models have theoretical or observational inconsistencies.
- B. ‘Oumuamua’s trajectory is inconsistent with a nongravitational acceleration that would be caused by the release of hydrogen gas resulting from the processing of water ice (H_2O), but the interstellar object’s observable properties can be explained if it has a significant component of molecular hydrogen ice (H_2).
- C. Since nongravitational accelerations of interstellar objects are several orders of magnitude weaker than gravitational accelerations, deviation from behavior that could be fully attributed to gravitational pull has been detected on a limited number of objects similar to ‘Oumuamua.

- D. Exposure to interstellar cosmic radiation can result in the formation of embedded pockets of hydrogen gas in water ice; moreover, when traveling through the solar system, ‘Oumuamua experienced warming sufficient to alter its icy structure and allow for outgassing.

Q5

INFORMATION AND IDEAS

Mosses can struggle in harsh desert conditions because these plants require enough sunlight for photosynthesis but not so much that they risk drying out. Researchers Jenna Ekwealor and Kirsten M. Fisher found several species of *Syntrichia caninervis*, a type of desert moss, growing under quartz crystals in California’s Mojave Desert. To evaluate whether these semitransparent rocks benefited the moss, the researchers compared the shoot tissue, a measure of plant growth, of *S. caninervis* when growing on the soil surface versus when the moss was growing under the quartz rocks. They found that the shoot tissue was 62% longer for moss growing under the quartz as compared to moss on the soil surface, suggesting that _____blank

Which choice most logically completes the text?

- A. *S. caninervis* is one of the few types of moss that can survive under semitransparent rocks.
- B. quartz crystals do not transmit the necessary sunlight for photosynthesis in *S. caninervis*.
- C. *S. caninervis* growing under quartz crystals experience lower light intensity and are thus able to retain more moisture.
- D. quartz crystals are capable of supporting *S. caninervis* growth if the crystals are not too thin.

The following text is Vita Sackville-West's circa 1920 poem "Evening." Spars are ships' masts, moorings are ropes that hold docked ships in place, and a riding-light is a light that a ship shines when it is anchored.

When little lights in little ports come out,
Quivering down through water with the stars,
And all the fishing fleet of slender spars
Range at their moorings, veer with tide about;

When race of wind is stilled and sails are furled,
And underneath our single riding-light
The curve of black-ribbed deck gleams palely white,
And slumbrous waters pool a slumbrous world;

—Then, and then only, have I thought how sweet
Old age might sink upon a windy youth,
Quiet beneath the riding-light of truth,
Weathered through storms, and gracious in retreat.

Which choice best states the main idea of the text?

- A. The tranquility of a port in the evening can incline a person to appreciate the stillness of old age.
- B. The difficulty of bringing a ship into port is apt training for dealing with the types of struggles encountered in old age.

- C. A person who leads a long life that is varied and active may find it difficult to stay in a calm place.
- D. The contrast between the peacefulness of a port at night and its activity during the day reflects the contrast between the calm of old age and the vibrancy of youth.

Percentage of Available Eggs Eaten by Cane Toad Tadpoles

Amphibian species (common name)	Percentage of eggs eaten	Native to Australia	Produces bufadienolide
Little red tree frog	1%	yes	no
Cane toad	90%	no	yes
Short-footed frog	7%	yes	no
Striped burrowing frog	10%	yes	no
Dainty green tree frog	1%	yes	no

Native to Latin America, the cane toad was introduced to Australia in the 1930s. In recent decades, tadpoles in the Australian population have been shown to consume eggs of their own species. A 2022 study showed that when presented with cane toad eggs as well as eggs of native Australian amphibians, cane toad tadpoles disproportionately consumed eggs of their own species. This behavior results from their attraction to bufadienolide, a chemical produced by the eggs of cane toads but not by the eggs of native amphibians. However, using data from this study, a student wishes to argue that the presence of bufadienolide doesn't entirely explain the cane toad tadpoles' preference for certain eggs over others.

Which choice best describes data from the table that support the student's argument?

- A. The tadpoles consumed a higher percentage of the striped burrowing frog eggs than they did of the eggs of the dainty green tree frog.
- B. The tadpoles left a certain percentage of the eggs of each of the five species unharmed, thus ultimately allowing them to hatch.
- C. The tadpoles consumed a lower percentage of the short-footed frog eggs than they did of the eggs of their own species.
- D. The tadpoles consumed the same percentage of the dainty green tree frog eggs as they did of the little red tree frog eggs.

In a three-year study of parasitic infections by *Anomotaenia brevis* tapeworms in *Temnothorax nylanderi* ants, entomologist Susanne Foitzik and colleagues found something unexpected: rather than reducing its host's fitness, as is typical of parasites, *A. brevis* greatly extends the lifespan of a *T. nylanderi* worker ant and seems to halt the effects of aging. Furthermore, those infected receive special treatment, ceasing their share of labor to sustain the colony and remaining in the nest as uninfected workers feed, groom, and transport them. By contrast, the researchers observed that uninfected workers in parasitized colonies have shortened lifespans, most likely because the _____ blank

Which choice most logically completes the text?

- A. uninfected workers are at high risk for direct exposure to *A. brevis* in the course of providing social care to the infected workers in the nest.
- B. need to compensate for reduced contributions within the colony while also caring for infected workers is burdensome to the uninfected workers.
- C. high level of activity maintained by the uninfected workers makes them better able than infected workers to quickly disperse when the nest is attacked by a predator.
- D. average lifespan of *T. nylanderi* worker ants in colonies without parasitic activity typically falls well below three years, the range covered by the study.

The most recent iteration of the immersive theater experience *Sleep No More*, which premiered in New York City in 2011, transforms its performance space—a five-story warehouse—into a 1930s-era hotel. Audience members, who wander through the labyrinthine venue at their own pace and follow the actors as they play out simultaneous, interweaving narrative loops, confront the impossibility of experiencing the production in its entirety. The play’s refusal of narrative coherence thus hinges on the sense of spatial fragmentation that the venue’s immense and intricate layout generates.

What does the text most strongly suggest about *Sleep No More*’s use of its performance space?

- A. The choice of a New York City venue likely enabled the play’s creators to experiment with the use of theatrical space in a way that venues from earlier productions could not.
- B. Audience members likely find the experience of the play disappointing because they generally cannot make their way through the entire venue.
- C. The production’s dependence on a particular performance environment would likely make it difficult to reproduce exactly in a different theatrical space.
- D. Audience members who navigate the space according to a recommended itinerary will likely have a better grasp of the play’s narrative than audience members who depart from that itinerary.

In a research paper, a student criticizes some historians of modern African politics, claiming that they have evaluated Patrice Lumumba, the first prime minister of what is now the Democratic Republic of the Congo, primarily as a symbol rather than in terms of his actions.

Which quotation from a work by a historian would best illustrate the student's claim?

- A. "Lumumba is a difficult figure to evaluate due to the starkly conflicting opinions he inspired during his life and continues to inspire today."
- B. "The available information makes it clear that Lumumba's political beliefs and values were largely consistent throughout his career."
- C. "Lumumba's practical accomplishments can be passed over quickly; it is mainly as the personification of Congolese independence that he warrants scholarly attention."
- D. "Many questions remain about Lumumba's ultimate vision for an independent Congo; without new evidence coming to light, these questions are likely to remain unanswered."